

Equality of Opportunity

Economics of Public and Social Issues

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Outline

1. Geographical Variation in Upward Mobility in America
2. Causal Effects of Places vs. Sorting

Lecture based primarily on the following paper:

Chetty, Friedman, Hendren, Jones, Porter. "The Opportunity Atlas: Mapping the Childhood Roots of Social Mobility" NBER WP, 2018

Quiz: AI and learning

Today's quiz

Three questions on [Strömberg, Lei and Wu \(2026\)](#), *The Generative AI Learning Penalty*.

- ▶ Answer in **iClicker** — scan the QR code on screen or open the app
- ▶ One answer per question; we look at the results together before moving on
- ▶ This is a **graded quiz**: you are scored on **getting it right**
- ▶ Quizzes are **12%** of your grade, and your **two lowest scores are dropped**

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Two of the three ask *how the study was done*, not just what it found — that is the part that will matter for your own projects.

Q1. Where do the data come from?

- A.** A survey in which students remembered their past grades
- B.** Administrative records from the county education bureau — the same digital systems that log exam scores, homework scores, and homework timestamps
- C.** An experiment in which researchers assigned some students to use AI
- D.** Test scores published online by the Chinese government

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Answer: B. 26,811 students — 90% of the county. The homework platform records *when* each assignment was opened and submitted, which is how the authors can spot homework that was finished very fast and scored very well.

Q2. Nobody was assigned to use AI. So how can the authors blame AI?

- A. They compare AI users with non-users at a single point in time
- B. They compare each student's scores **before vs. after she started using AI**, then subtract whatever happened over those same months to students who **never** started
- C. They asked students whether they felt AI had hurt their grades
- D. They compare schools that banned AI with schools that allowed it

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Answer: B. Why subtract the never-users? Exams get harder, seasons change, students mature — the never-users soak all of that up. And the two groups moved **in parallel before** anyone adopted, which is what makes the comparison believable.

Q3. What happened after students started using AI?

- A. Homework scores and exam scores both went **up**
- B. Homework scores and exam scores both went **down**
- C. Homework scores went **up** and homework took **less time**, but exam scores went **down**
- D. Homework was unchanged, but exam scores went **up**

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Answer: C. Homework scores **+18%**, time per assignment **64 → 45 minutes**, closed-book exam scores **−20%**. The work got easier. The learning did not happen.

So what?

- ▶ The students who lost the most were the ones who used AI to **finish faster**
- ▶ The students who used AI **and kept their time on the task** scored about the same as everyone else
- ▶ Nobody here was punished for *using a tool* — they were punished for *skipping the thinking*

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Keep this in mind on Project 0: the goal is not a finished file, it is being able to **explain** the file.

Geographical Variation in Upward Mobility

Differences in Opportunity Across Local Areas

- ▶ How do children's chances of moving up vary across areas in America?
 - ▶ Are there some areas where kids do better than others? If so, what lessons can we learn from them?
- ▶ Recent studies have used big data to measure how upward mobility varies based on where children grow up

The Opportunity Atlas: Data Sources and Sample Definitions

- ▶ Data sources: Anonymized Census data (2000, 2010, ACS) covering U.S. population linked to federal income tax returns from 1989-2015
- ▶ Link children to parents based on dependent claiming on tax returns
- ▶ Target sample: Children in 1978-83 birth cohorts who were born in the U.S. or are authorized immigrants who came to the U.S. in childhood
- ▶ Analysis sample: 20.5 million children, 96% coverage rate of target sample

Measuring Parents' and Children's Incomes in Tax Data

- ▶ Parents' household incomes: average income reported on Form 1040 tax return from 1994-2000
- ▶ Children's incomes measured from tax returns in 2014-15 (ages 31-37)
- ▶ Focus on percentile ranks in national distribution:
- ▶ Rank children relative to others born in the same year and parents relative to other parents

Intergenerational Income Mobility for Children Raised in Chicago

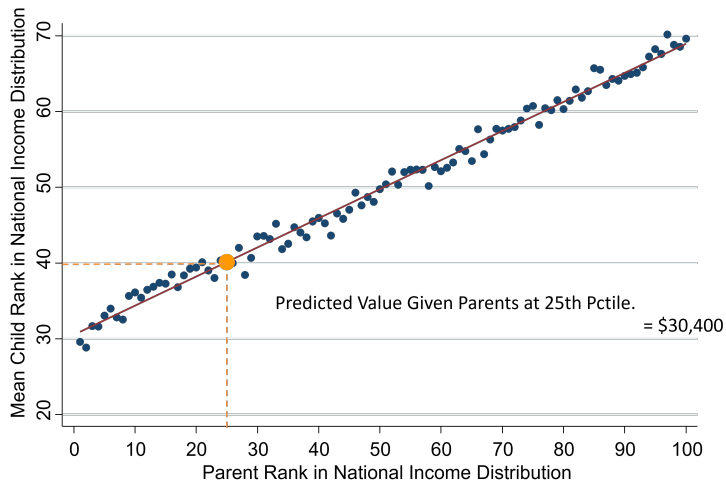


Figure 1: Source: [Chetty, Hendren, Kline, Saez \(QJE 2014\)](#)

Intergen. Mobility for Children Raised in Hypothetical Census Tract

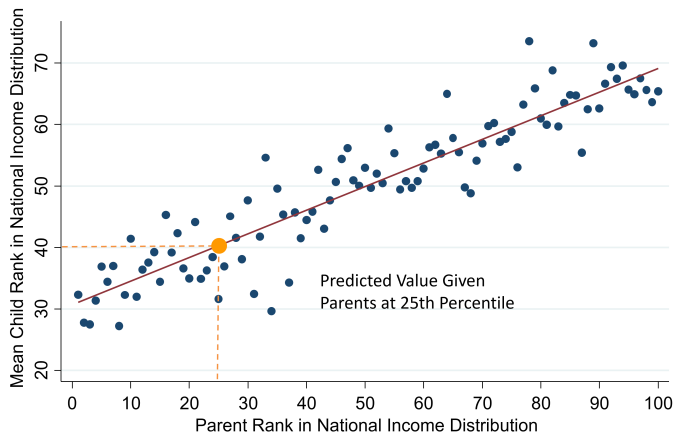


Figure 2: Source: [Chetty, Hendren, Kline, Saez \(QJE 2014\)](#)

Estimating Children's Average Outcomes by Census Tract

- ▶ Run a separate regression using data for children who grow up in each Census tract in America
- ▶ In practice, many children move across areas in childhood
- ▶ Weight children by fraction of childhood (up to age 23) spent in a given area

The Geography of Upward Mobility in the United States

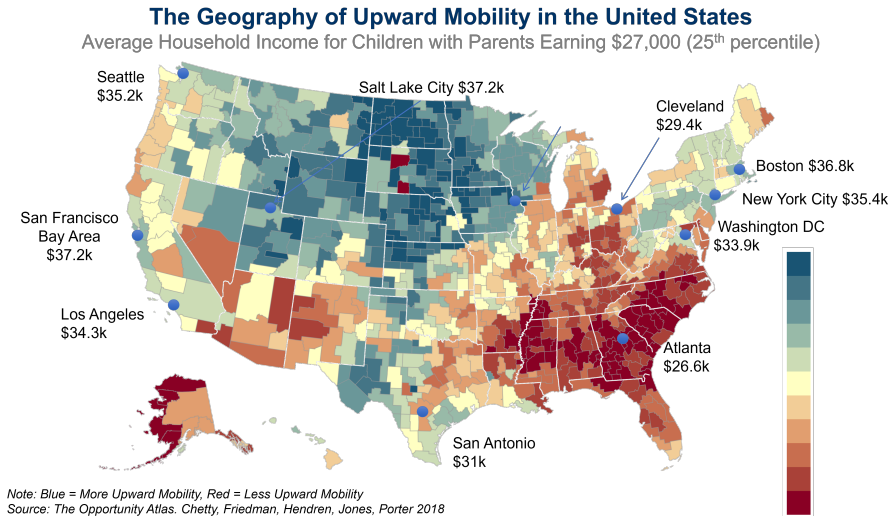


Figure 3: Source: Chetty, Friedman, Hendren, Jones and Porter (2018)

Jump to Opportunity Atlas website

- ▶ <https://www.opportunityatlas.org/>
- ▶ Explore Durham and San Antonio briefly